Introduction to Biomaterials

Lecture [5]

Polymeric Biomaterials: Structure, Properties, and Biomedical Applications

Goals

- Structure of polymers and their common physical states.
- General properties of polymers.
- Common types of polymerization techniques used for production.
- Chemical structure and properties of common biomedical polymers.
- Structure of hydrogels and their general properties.
- Definition and uses of nanopolymers.

Polymers as Biomaterials

- Polymers can be **natural** or **synthetic** materials.
- Most polymers are **carbon-based**.
- Common examples include **polyethylene**, **PMMA**, **polystyrene**, **starch**, **cellulose**, and **rubber**.
- Many biomedical polymers are chemically similar to **everyday plastics**, but must meet strict medical safety standards.
- The use of polymers as biomaterials has increased significantly over the past **75 years** as advances in polymer science have yielded a variety of polymers.

Common Biomedical Polymer Examples

- UHMWPE is used in **total joint prostheses**.
- PMMA is used as **bone cement** in orthopedic surgery.

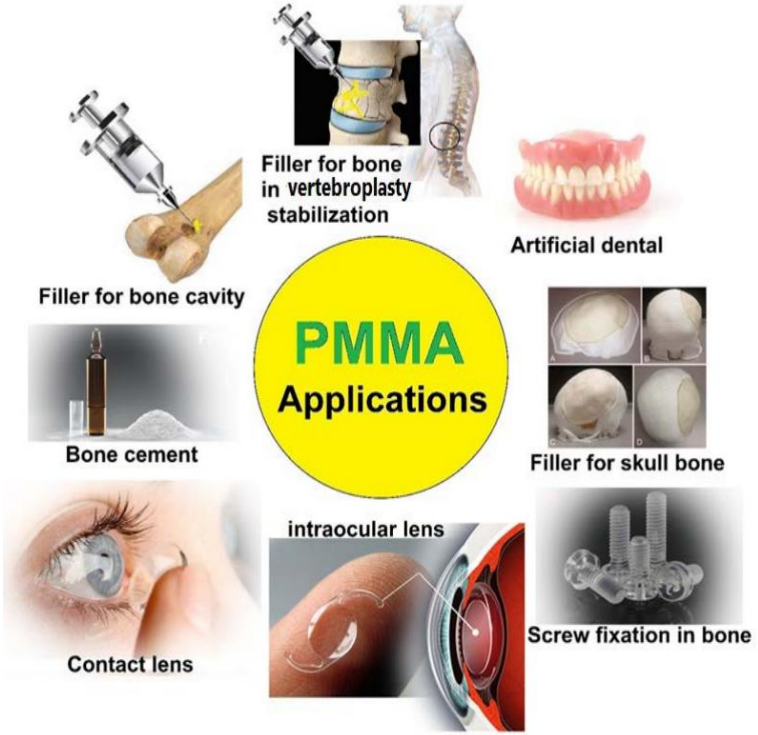
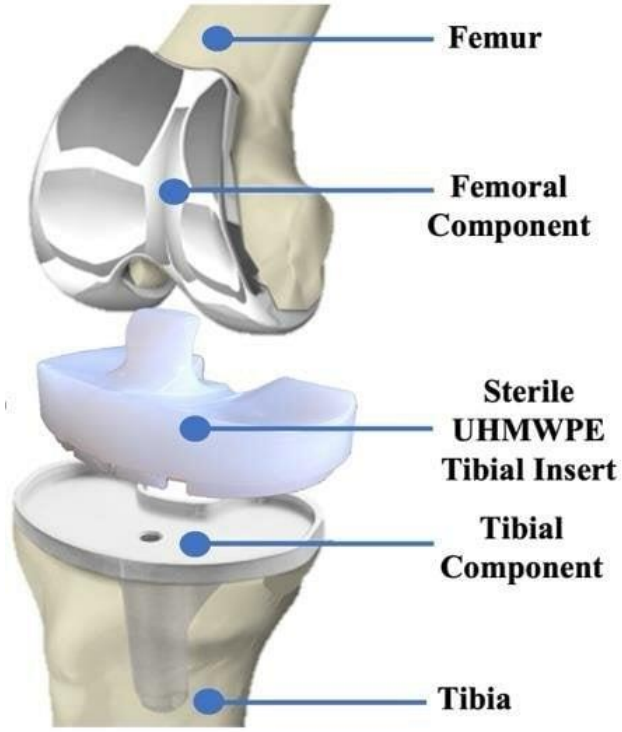


Fig. 4. Medical applications of PMMA

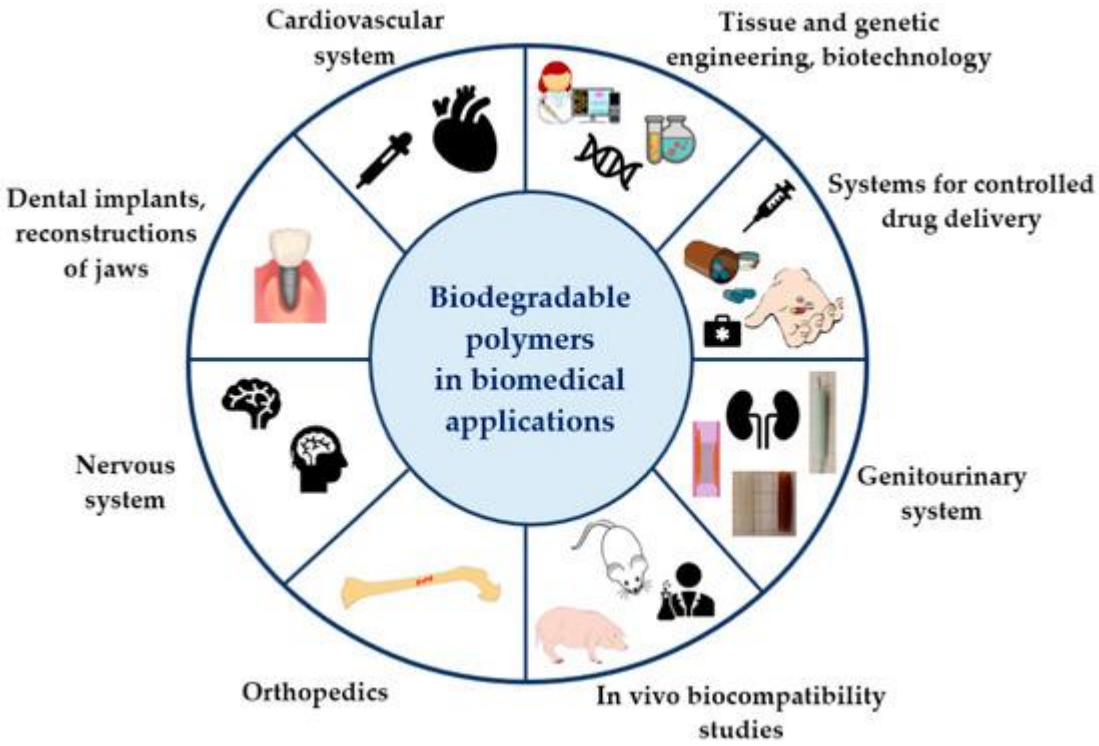


Advantages of Polymers in Biomaterials

- Polymers are **easy to shape** using **molding**, **casting**, **machining**, or **other** methods.
- Polymer implants are often **less expensive** to manufacture.
- Their surfaces can be chemically modified to **improve compatibility** with body tissues.
- Some polymers can be designed to **degrade inside the body**.

Biodegradable Polymers

- **Biodegradable polymers** can gradually break down after completing their function.
- They are useful for temporary implants.
- They may reduce the need for a second surgery.
- They can also be designed to release drugs or growth factors during degradation.

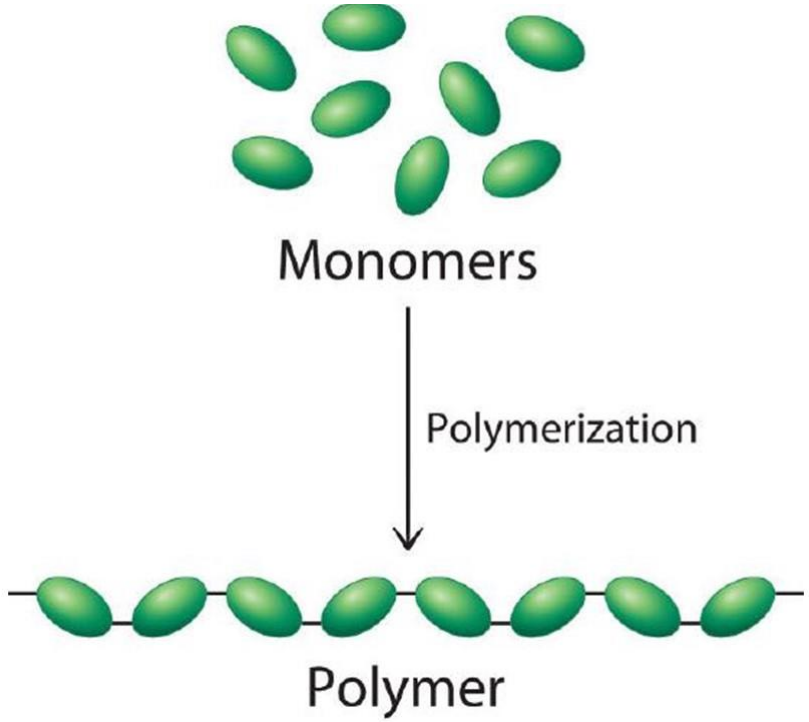


Limitations of Polymers

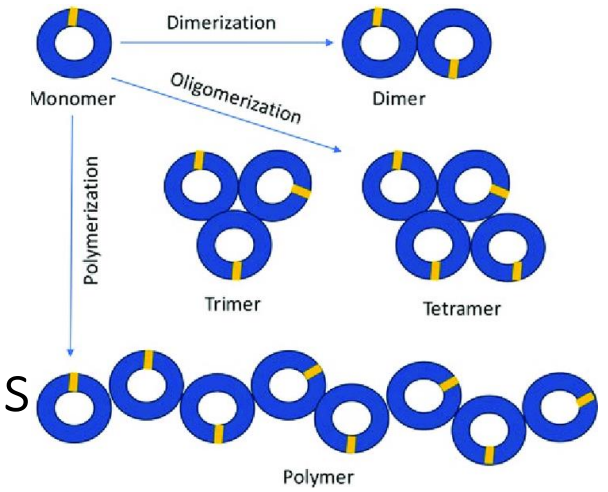
- Polymers are usually **weaker** and **less stiff** than **metals** or **ceramics**.
- They may **not be suitable** for implants carrying **high mechanical loads**.
- Polymer selection depends strongly on the required function inside the body.
- Mechanical strength, degradation, and biocompatibility must be carefully considered.

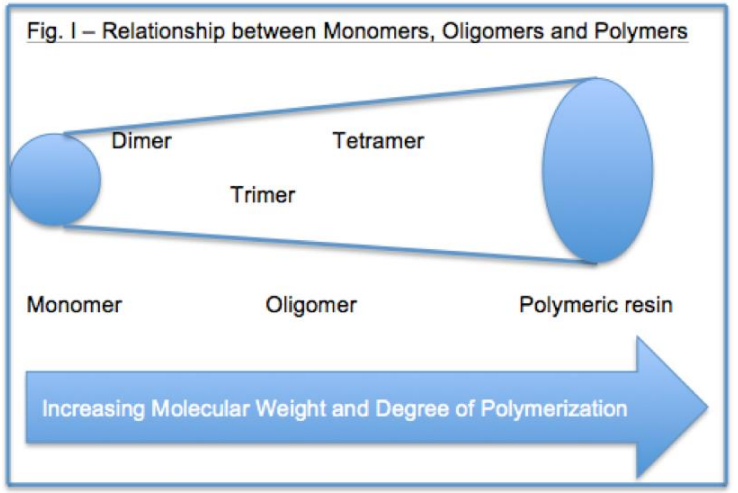
- Polymers **are long-chain molecules made of repeated units called monomers**.

- The **word polymer** means **many parts**.
- A polymer chain may contain hundreds, thousands, or millions of repeat units.
- **Short chains** with only a **few repeat units** are called **oligomers**.



Molecular Structure of Polymers

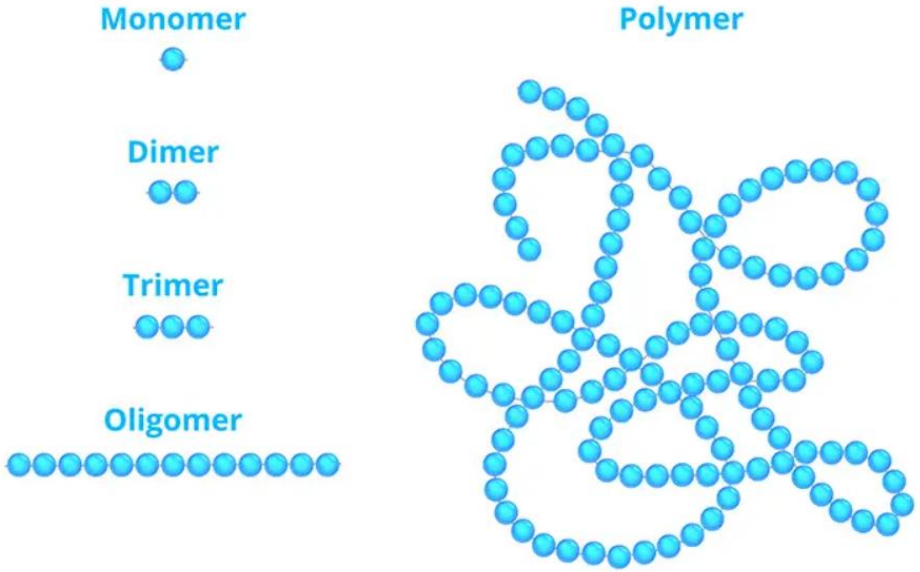




- **Monomer** = one unit
  Example: A
- **Oligomer** = few repeated units
  Example: A–A–A–A
  Usually has **low molecular weight** and short chain length.
- **Polymer** = many repeated units
  Example: A–A–A–A–A–A–A–A–A–A–…
  Usually has **high molecular weight** and long chains.

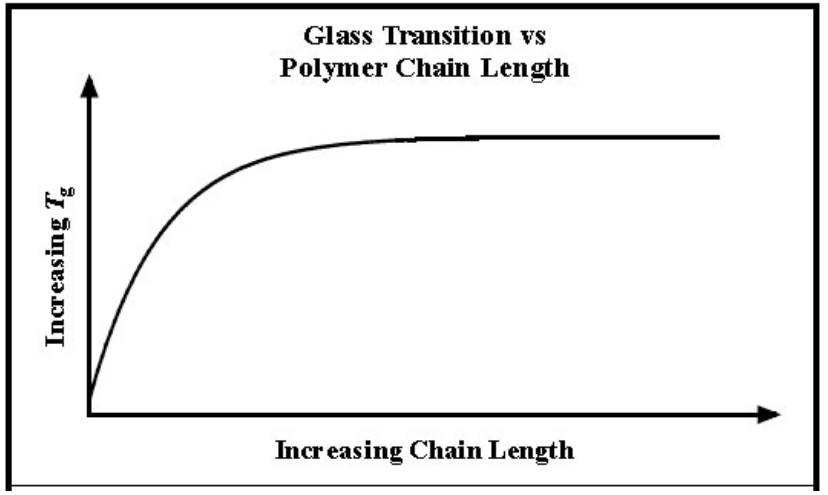
Monomers, Repeat Units, and End Groups

- **Monomer:** small molecule used to build the polymer.
- **Repeat unit:** repeating chemical unit inside the polymer chain.
- **End groups:** chemical groups located at the ends of the polymer chain.
- Polymer **properties** depend strongly on **chain length** and **molecular structure**.



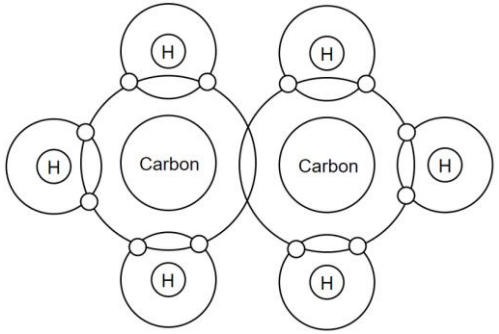
Effect of Chain Length on Polymer Properties

- Increasing chain length changes material behavior.
- **Short** molecules may behave like **gases** or **liquids**.
- **Longer** chains can form **waxy** or **solid materials**.
- **Very long** chains can entangle, giving **typical polymer properties**.
- Example: polyethylene properties change with molecular weight.



Bonding in Polymers

- Atoms inside one polymer chain are mainly connected by **covalent bonds**.
- **Covalent bonds strongly** affect **mechanical**, **thermal**, and **chemical** properties.
- Neighboring chains interact through weaker secondary bonds.
- **Secondary bonds** affect **melting**, **dissolution**, **diffusion**, and **flow behavior**.



Homopolymers & Copolymers

- **Homopolymer**: made from one type of monomer.
- **Copolymer**: made from two or more types of monomers.
- Copolymer structure affects final polymer properties.
- Changing **monomer type** or **arrangement** can change **strength**, **flexibility**, and **degradation** behavior.



Figure 6.1

Homopolymer with A monomers.

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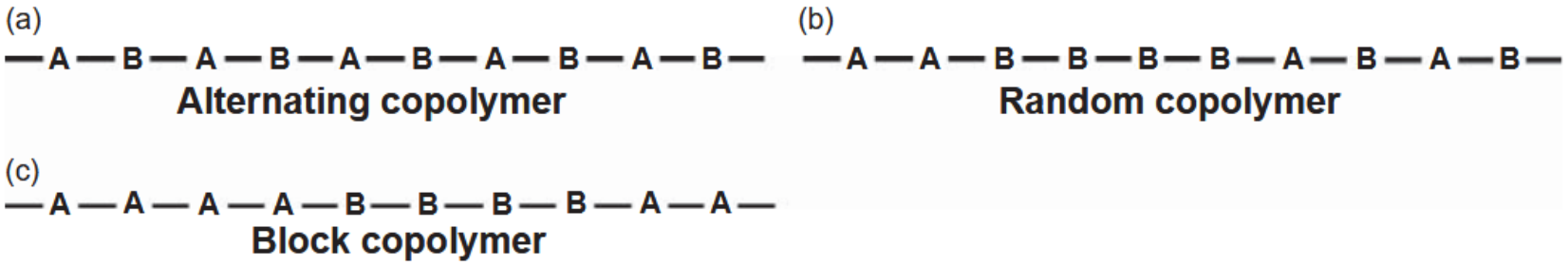


Figure 6.2

Polymerization of A and B monomers to form (a) alternate copolymers, (b) random copolymers, and (c) block copolymers.

Monomer chemical structure/formula and the polymer repeating unit.

Monomer	Monomer structure / composition	Polymer formed	Polymer repeating unit
Ethylene / Ethene	**$CH_2=CH_2$**	Polyethylene **PE**	**$-[CH_2-CH_2]_n-$**
Propylene / Propene	**$CH_2=CH-CH_3$**	Polypropylene **PP**	**$-[CH_2-CH(CH_3)]_n-$**
Vinyl chloride	**$CH_2=CHCl$**	Polyvinyl chloride **PVC**	**$-[CH_2-CHCl]_n-$**
Styrene	**$CH_2=CH-C_6H_5$**	Polystyrene **PS**	**$-[CH_2-CH(C_6H_5)]_n-$**
Lactic acid	**$CH_3-CH(OH)-COOH$**	Polylactic acid **PLA**	**$-[O-CH(CH_3)-CO]_n-$**
Lactide	**$C_6H_8O_4$** cyclic dimer of lactic acid	Polylactic acid **PLA**	**$-[O-CH(CH_3)-CO]_n-$**

Types of Copolymers

- **Alternating copolymer:** monomers repeat in regular A–B pattern.
- **Random copolymer:** monomers are arranged randomly.
- **Block copolymer:** long blocks of each monomer are connected.
- Different arrangements produce different material properties.

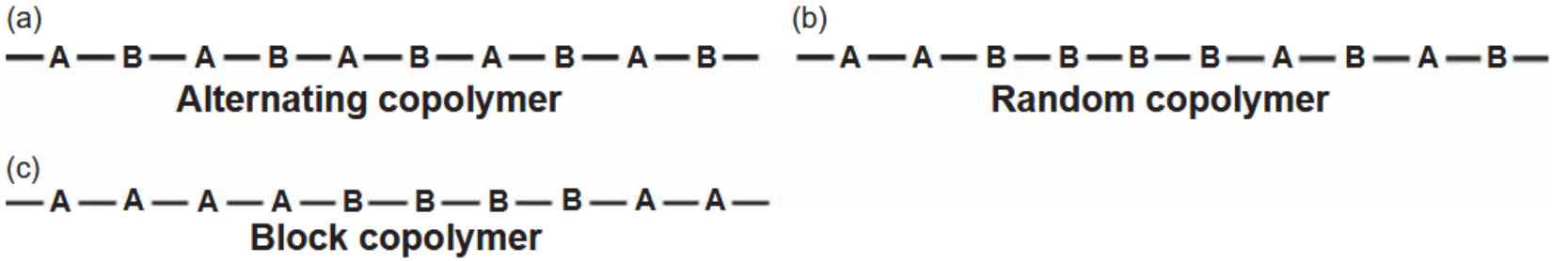
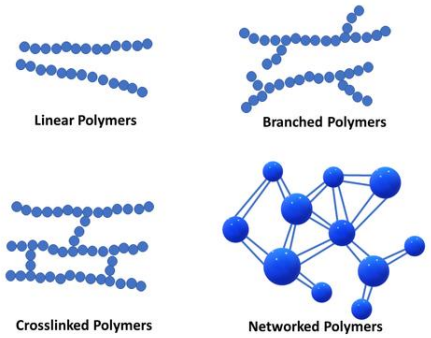


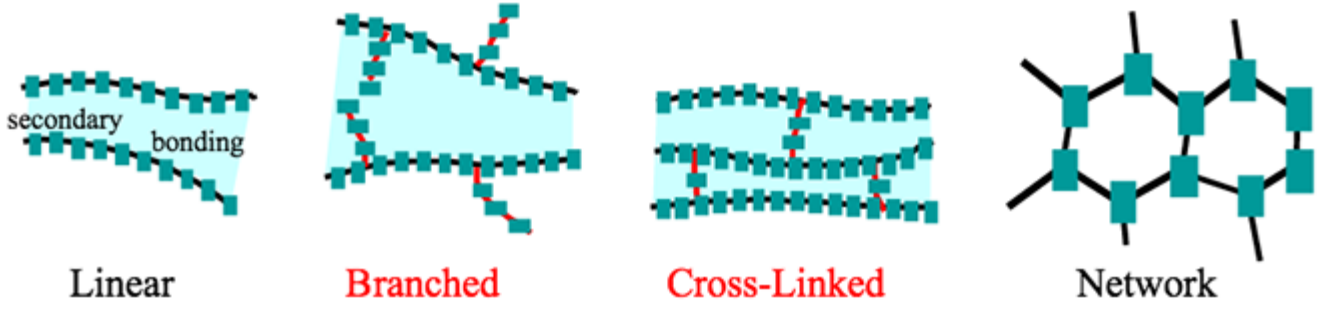
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Polymer Chain Architectures

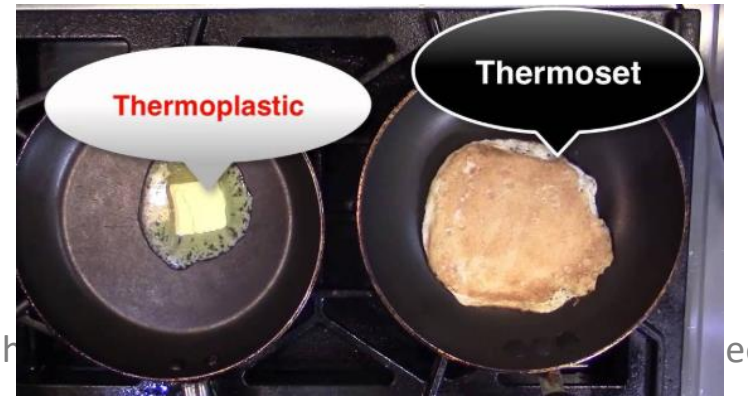
- Polymer chains can be **linear**, **branched**, or **networked**.
- **Linear polymers:** simple chain-like structure.
- **Branched polymers:** side chains attached to main backbone.
- **Networked polymers:** chains connected together by crosslinks.
- **Chain architecture** strongly affects **flexibility**, **melting**, and **mechanical** behavior.

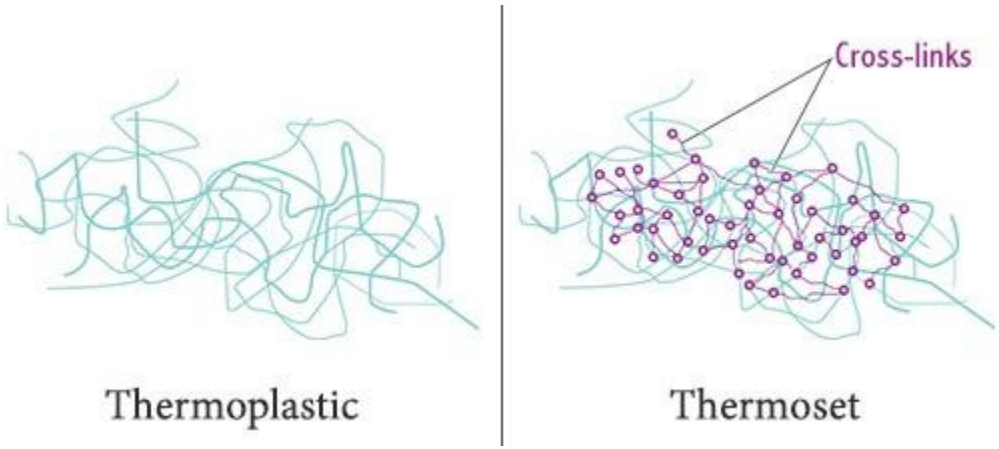




Thermoplastics and Thermosets

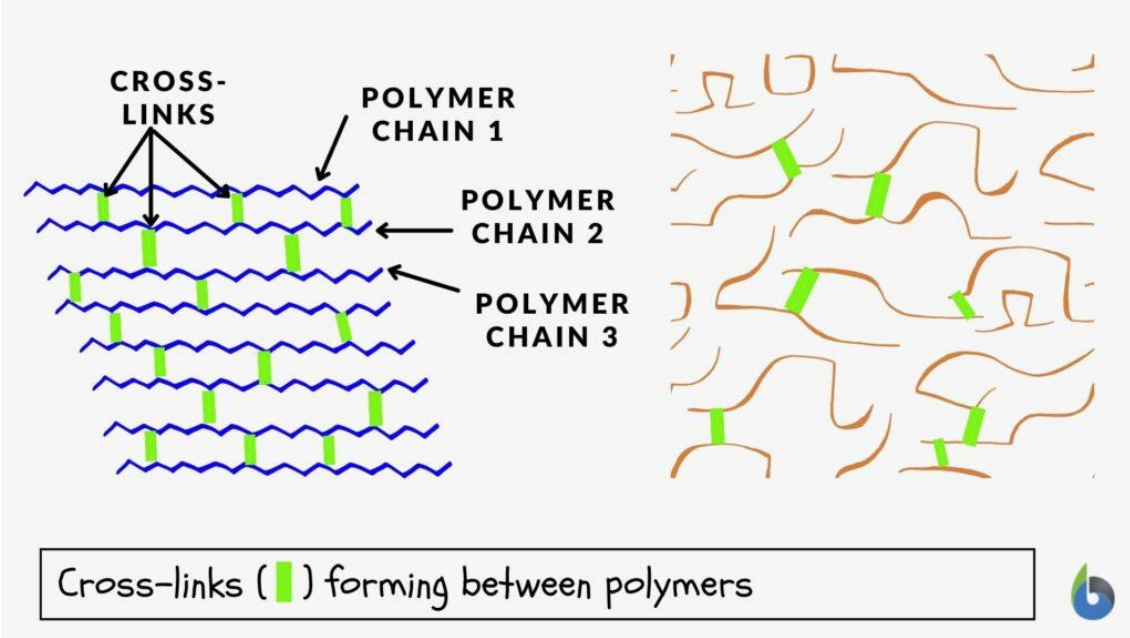
- **Thermoplastics:** can soften or melt when heated and can be reshaped.
- Usually have linear or branched chains.
- **Thermosets:** crosslinked polymers that do not melt or reshape easily.
- **Crosslinking** makes polymers more rigid and dimensionally stable.
- **Biomedical** processing depends strongly on whether the polymer is **thermoplastic** or **thermoset**.





Crosslinking in Polymers

- **Crosslinking** connects polymer chains into a network.
- **More crosslinking reduces chain movement.**
- **Crosslinked** polymers become **stiffer and more stable**.
- **Crosslinking** can occur during **polymerization** or **after processing**.
- **Radiation** or **heat** can sometimes **create crosslinks**.



Polymer Tacticity

- Tacticity describes the arrangement of side groups along the polymer chain.
- **Isotactic**: side groups on the same side.
- **Syndiotactic**: side groups alternate sides.
- **Atactic**: side groups arranged randomly.
- **Tacticity** affects crystallinity and material properties.

Polymer Tacticity

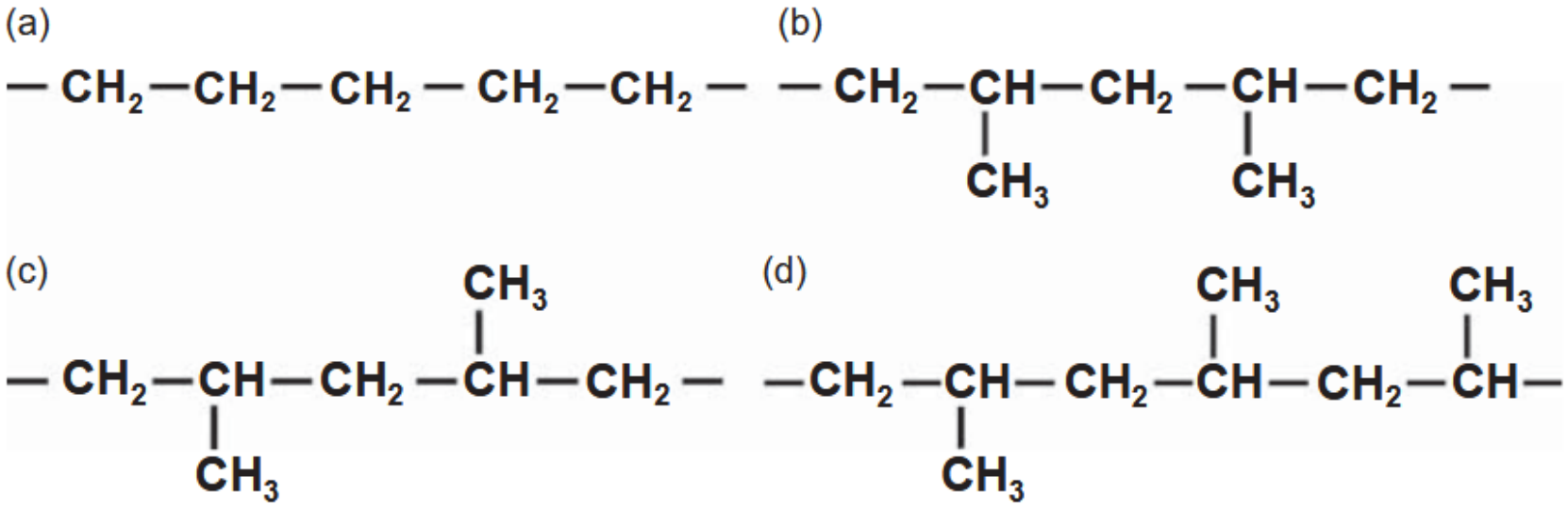


Figure 6.5

Different forms for a polymeric molecular chain. (a) A polymer with no asymmetry, (b) isotactic – each pendant group is on the same side of the chain, (c) syndiotactic – the pendant groups appear on alternate sides of the chain in a regular fashion, and (d) atactic – the pendant groups appear on opposite sides of the chain in a random fashion.

Why Molecular Structure Matters in Biomaterials

- Polymer structure controls mechanical strength, flexibility, and stability.
- Chain length, branching, crosslinking, and monomer arrangement affect performance.
- These structural features influence degradation, processing, and tissue interaction.
- Understanding structure helps engineers choose suitable polymers for medical devices and implants.

Molecular Weight of Polymers

- Polymer chains do not all have the same length.
- A polymer sample contains chains with different numbers of repeat units.
- Therefore, polymer molecular weight is usually expressed as an average value.
- Molecular weight strongly affects polymer properties and processing behavior.

Molecular Weight Distribution

- Polymer samples contain a distribution of chain lengths.
- Chains may range from short oligomers to very long polymer chains.
- The width of this distribution depends on polymerization conditions.
- Controlling molecular weight distribution is important for predictable material performance.

Number Average and Weight Average Molecular Weight

- Number average molecular weight (Mn): more affected by short chains.
- Weight average molecular weight (Mw): more affected by long chains.
- Both values describe the polymer, but they give different information.
- Longer chains usually have stronger influence on mechanical and flow behavior.

$$M_n = \frac{\sum_{i=1}^{\infty} W_i}{\sum_{i=1}^{\infty} N_i} = \frac{\sum_{i=1}^{\infty} M_i N_i}{\sum_{i=1}^{\infty} N_i},$$

$$M_w = \frac{\sum_{i=1}^{\infty} M_i W_i}{\sum_{i=1}^{\infty} M_i N_i} = \frac{\sum_{i=1}^{\infty} M_i^2 N_i}{\sum_{i=1}^{\infty} M_i N_i}.$$

where

W_i = weight of chains in fraction i,
M_i = molecular weight of chains in fraction i, and
N_i = number of chains in fraction i.

Polydispersity Index

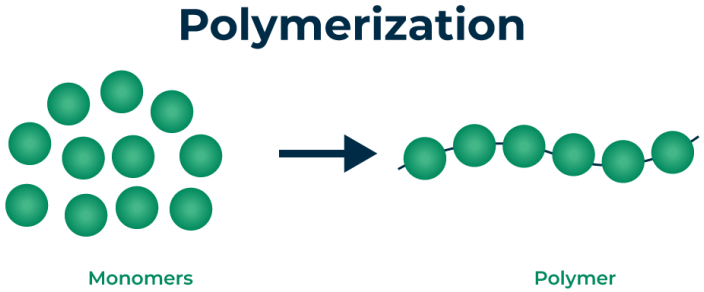
- The ratio **Mw/Mn** is called the **polydispersity index (PDI).**
- **PDI** shows **how wide the molecular weight distribution is**.
- **Low PDI** means **chains are more similar in length**.
- **High PDI** means the polymer contains a **wider range of chain lengths**.

Molecular Weight and Polymer Processing – example !

- Higher molecular weight usually increases viscosity.
- High-viscosity polymers are harder to melt, mold, or process.
- Example: UHMWPE has very high molecular weight.
- UHMWPE is useful in joint implants, but it is difficult to melt mold.



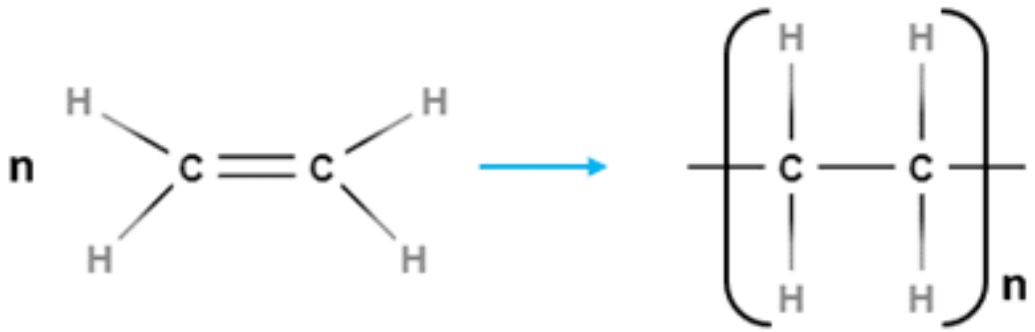
Polymerization



- Polymerization is the process of joining monomers into long polymer chains.
- Monomers are connected mainly by covalent bonds.
- The type of polymerization affects polymer structure, molecular weight, and final properties.
- Two main reaction types: addition polymerization and condensation polymerization.
- *Addition polymerization is "joining without loss," while condensation polymerization is "joining with release of a small molecule, usually water."*

Addition Polymerization

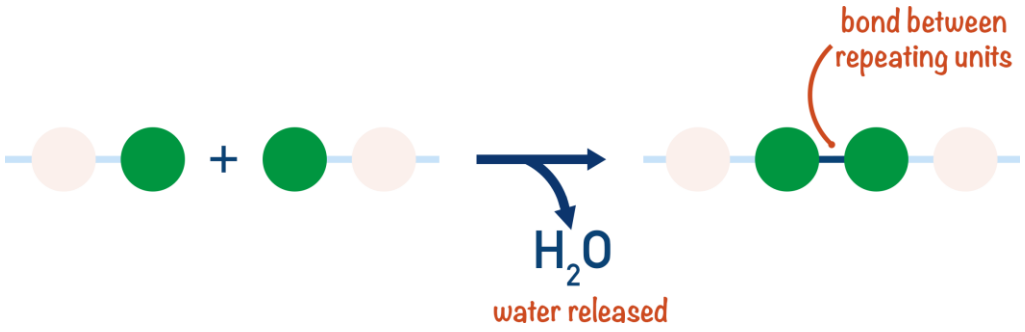
- Monomers with double bonds join together to form a growing polymer chain.
- No small molecule is usually lost during the reaction.
- The repeat unit has the same atoms as the original monomer.
- Common in polymers such as polyethylene and other vinyl-based polymers.

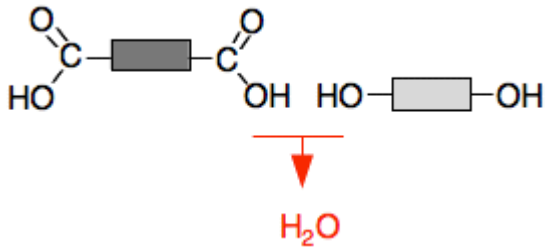


Ethene Poly(ethene)

Condensation Polymerization

- Monomers react together while releasing small by-products.
- The repeat unit may contain fewer atoms than the original monomer.
- Common by-products may include small molecules such as water.
- Used to form many important biomedical polymers.





Degree of Polymerization

- Degree of polymerization DP means the average number of repeat units in a polymer chain.
- Higher DP usually means longer chains and higher molecular weight.
- Molecular weight depends on the repeat unit mass and number of repeat units.
- DP is important because chain length affects strength, viscosity, degradation, and processing.

Why Polymerization Matters in Biomaterials

- Polymerization controls chain length, molecular weight, and polymer structure.
- These factors affect flexibility, strength, degradation rate, and biocompatibility.
- Proper control of polymerization is important for medical devices, implants, sutures, scaffolds, and drug delivery systems.

Physical States of Polymers

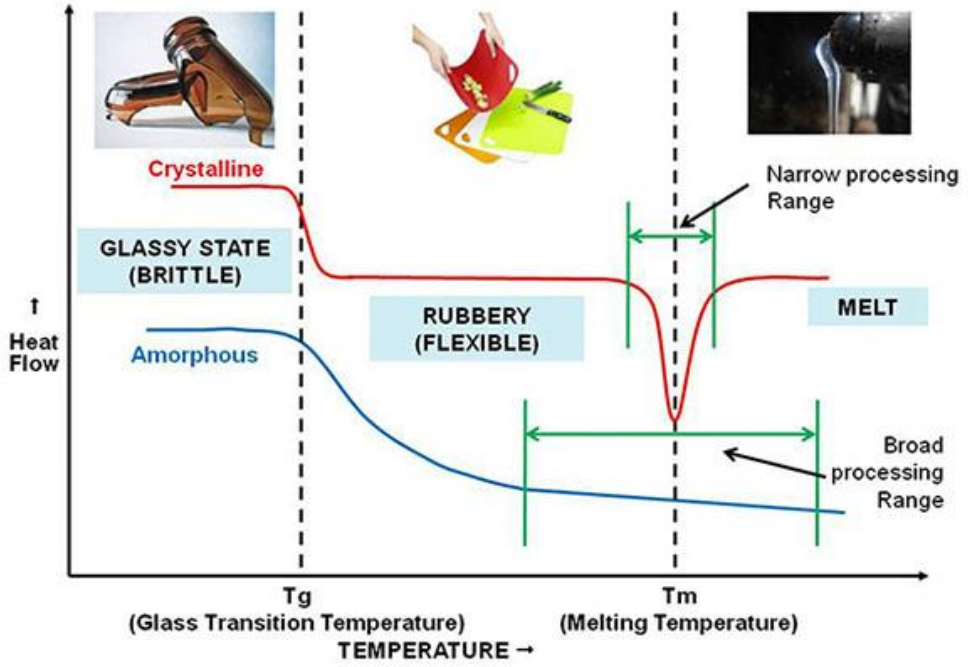
- Polymer chains may be random or ordered.
- Amorphous: random chains → more flexible.
- Crystalline: ordered chains → more stiff and strong.
- Semi-crystalline: contains both amorphous and crystalline regions.
- Tg: hard/glassy → soft/rubbery.
- Tm: crystalline parts melt and polymer can flow.

Physical States of Polymers - Tg

- **Tg = Glass Transition Temperature**
- It means the temperature where a polymer changes from:
- **hard / glassy / brittle → soft / rubbery / flexible**
- **Tg is not melting.**
  The polymer does **not** become liquid; it only becomes softer because the chains start moving more.
- Example:
- Below Tg: hard plastic
  Above Tg: rubber-like flexible material

Physical States of Polymers - Tm

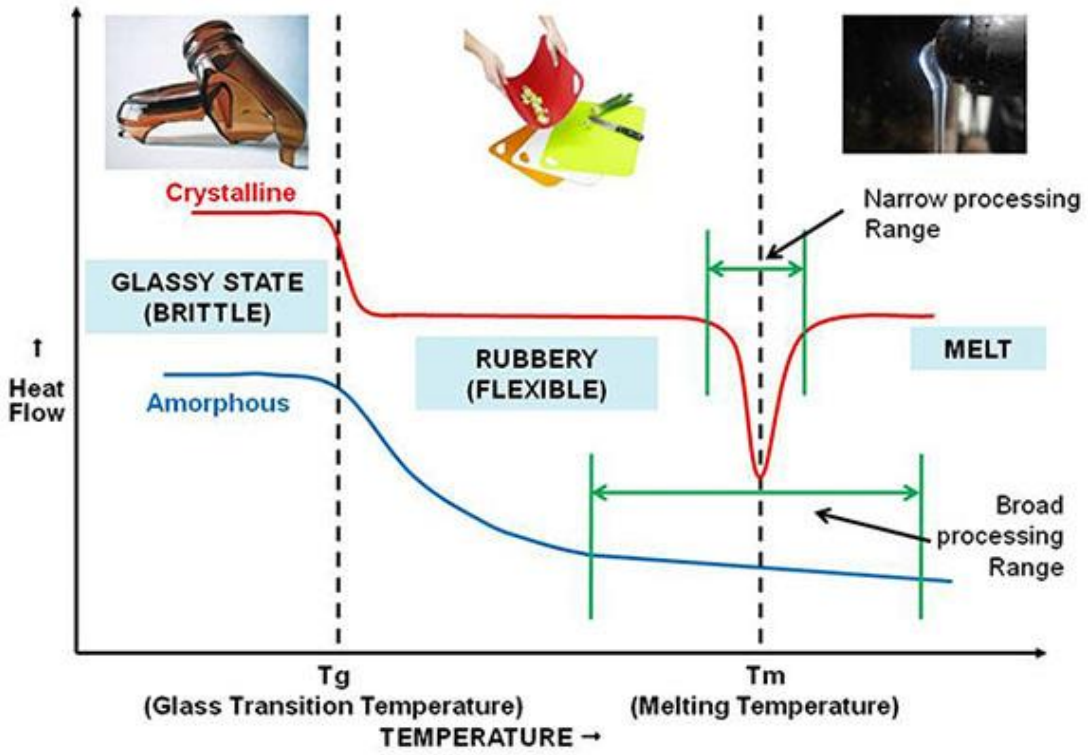
- Tm = Melting Temperature
- It means the temperature where the **crystalline/ordered parts** of a polymer melt.
- At **Tm**, the polymer can start to become a melt and flow.
- Example:
- Below Tm: solid polymer
  Above Tm: melted polymer



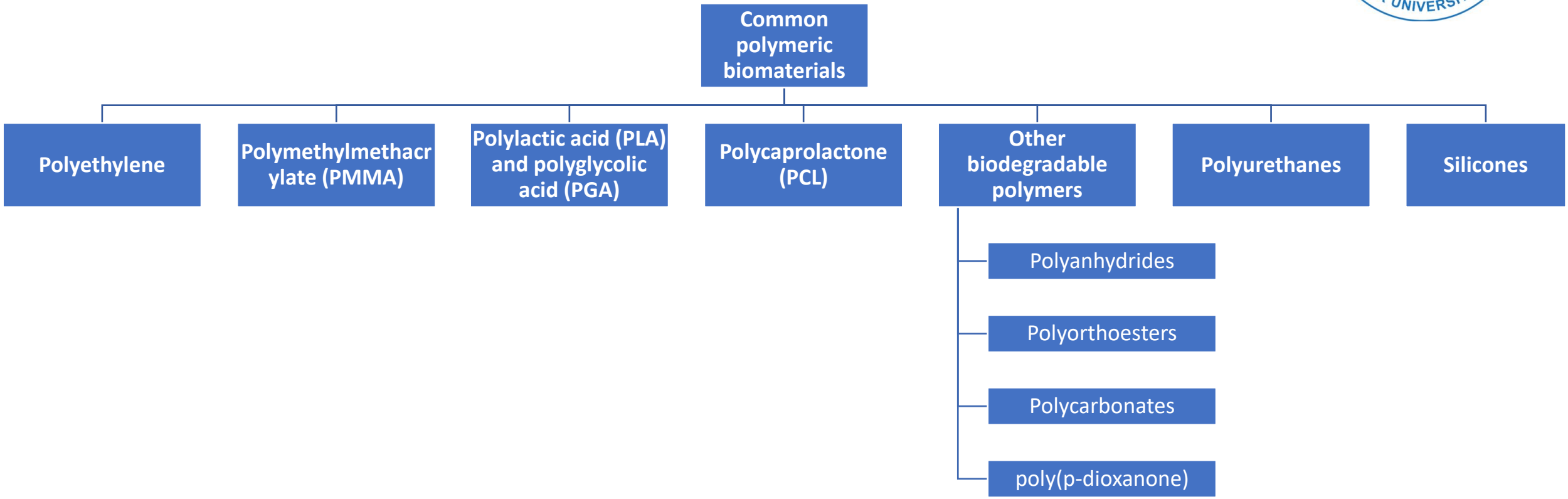
Physical States of Polymers – Tg / Tm

Amorphous polymers mainly have Tg. Semi-crystalline polymers can have both Tg and Tm.

Term	Meaning	Simple idea
Tg	Glass transition temperature	hard → soft
Tm	Melting temperature	solid → melt

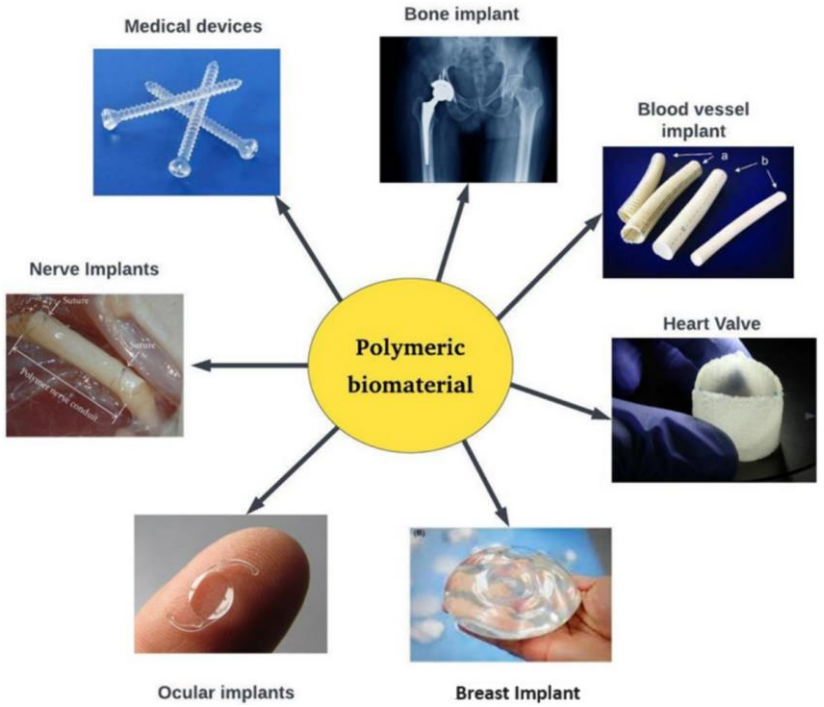


Common Polymeric Biomaterials


Common polymeric biomaterials
Polyethylene
Polymethylmethacrylate (PMMA)
Polylactic acid (PLA) and polyglycolic acid (PGA)
Polycaprolactone (PCL)
Other biodegradable polymers
Polyurethanes
Silicones
Polyanhydrides
Polyorthoesters
Polycarbonates
poly(p-dioxanone)

Common Polymeric Biomaterials

- Common biomedical polymers include **UHMWPE**, **PMMA**, **PLA**, **PGA**, **PCL**, **polyurethanes**, and **silicones**.
- They are used in **implants**, **prostheses**, **sutures**, **scaffolds**, **catheters**, and **drug delivery systems**.
- Polymer choice depends on mechanical properties, degradation behavior, biocompatibility, and processing method.



1) UHMWPE - in Joint Implants

- **Ultra-High-Molecular-Weight Polyethylene (UHMWPE)** is widely used as a **bearing material** in total joint prostheses.
- It has good wear resistance compared with many polymers.
- Main problem: it can produce very small wear particles.
- Wear debris can lead to inflammation, osteolysis, and implant loosening.





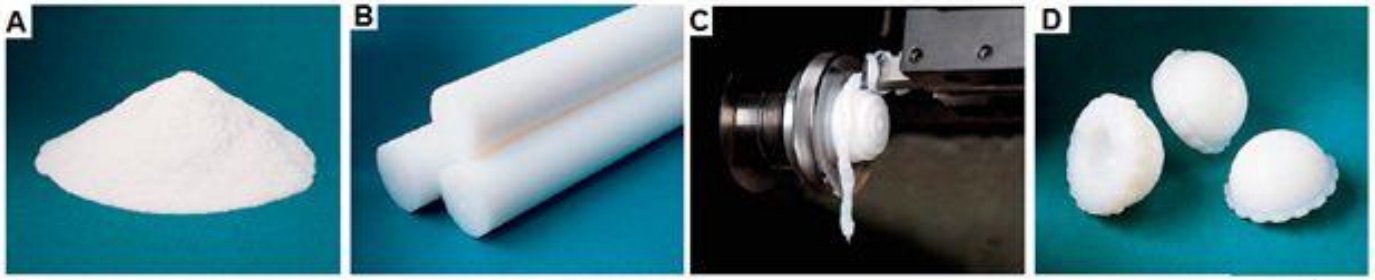
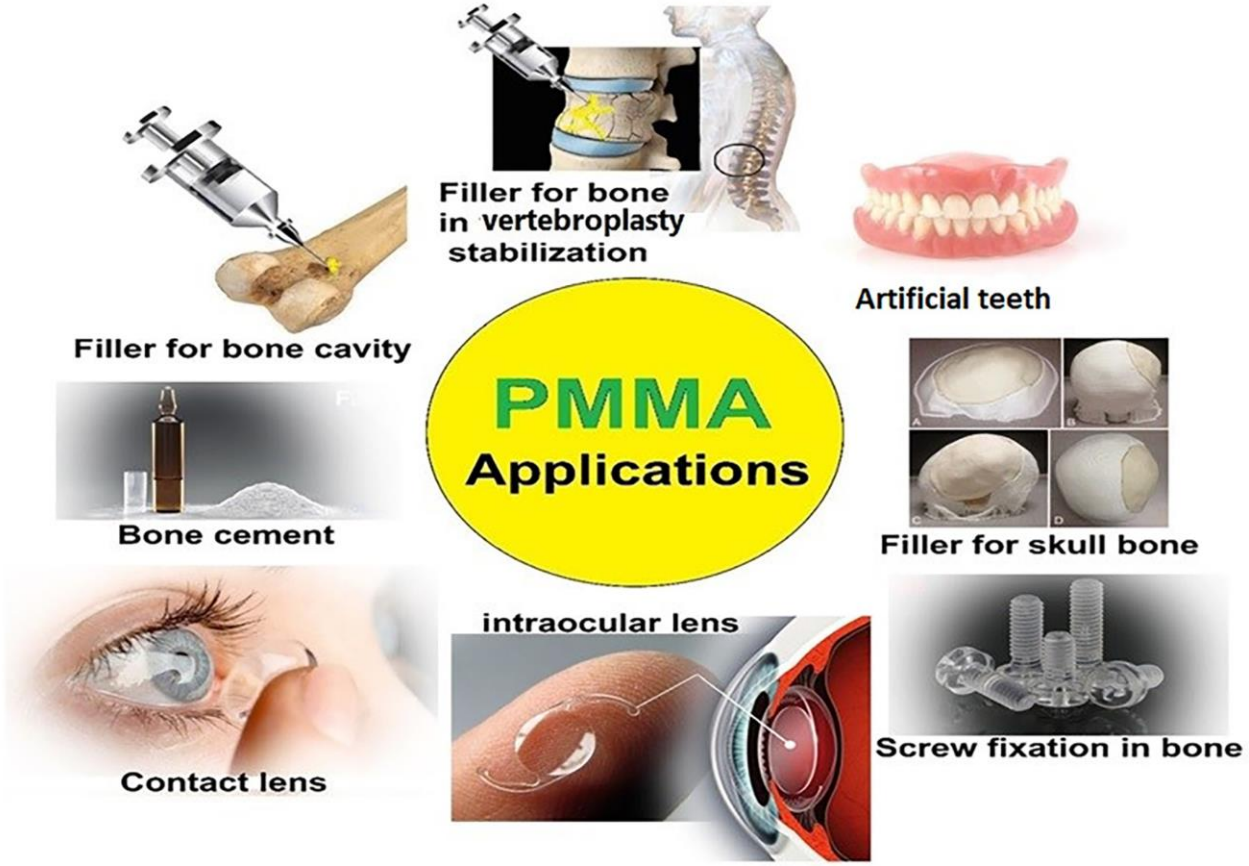


Figure 3: Typical processing steps in the manufacture of UHMWPE implants, starting with the resin powder (A). (B) Semi-finished rods that have been consolidated from the resin powder; (C) Machining of the UHMWPE rods on a lathe; (D) UHMWPE acetabular components after machining.

2) PMMA - Bone Cement

- **PMMA** is widely used as **bone cement** in **orthopedic surgery**.
- It helps stabilize total joint prostheses.
- It can also be used as a **bone substitute** in some fractures.
- PMMA used clinically is **chemically similar to Plexiglass**.

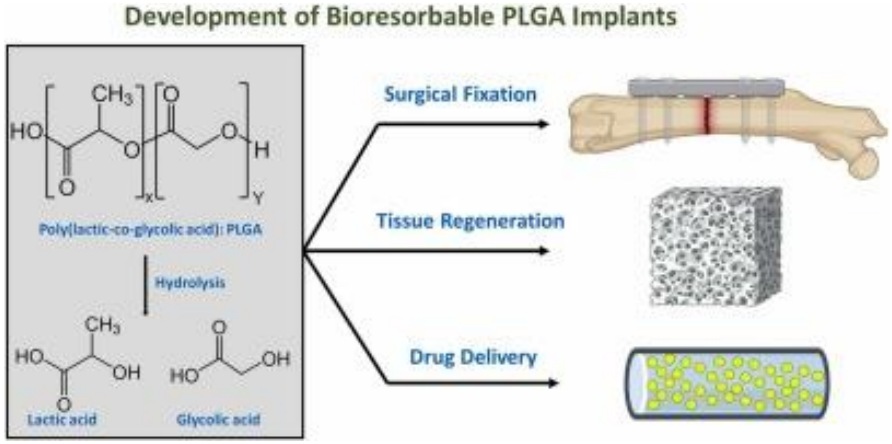


Bone Cement VS Bone Substitute

Feature	Bone Cement	Bone Substitute
Primary Goal	Immediate mechanical support & fixation	Bone regeneration & remodeling
Chemical Makeup	**PMMA** (Polymethylmethacrylate)	**Hydroxyapatite, TCP, $CaSO_4$** (similar to bone mineral)
Mode of Action	A **"grout"** – fills space for mechanical fixation	A **"scaffold"** – provides a framework for new bone
Strength	**High** – provides immediate strong fixation	**Moderate** – typically less than natural bone
Biological Role	**Non-bioactive** – remains as a permanent synthetic spacer	**Bioactive** – is remodeled and replaced by living bone
Long-term Fate	**Remains permanent** in the body as a synthetic cement	**Is resorbed** and replaced by natural bone over months to years
Drawback	Can loosen over time; exothermic (heat) reaction can damage tissue	Weaker initial strength; resorption rate must match patient's healing
Common Uses	Joint replacements (hip/knee), spinal fracture fixation (kyphoplasty)	Filling bone defects/cysts, spinal fusion, dental bone grafts

3) Polylactic acid (PLA) and polyglycolic acid (PGA)

- **PLA** and **PGA** are biodegradable polymers.
- They are linear polyesters used in many medical applications.
- Common uses: **biodegradable screws**, **plates**, **anchors**, **sutures**, **scaffolds**, and **drug delivery.**
- Their degradation occurs mainly by hydrolysis of ester bonds.



PLA/PGA Degradation

- **PGA degrades** to **glycolic acid** and then **enters natural body pathways**.
- **PLA degrades** to **lactic acid**, which **naturally occurs in the body**.
- **Final degradation** products can be converted to **water** and **carbon dioxide**.
- Degradation rate depends on crystallinity, molecular weight, and copolymer ratio.

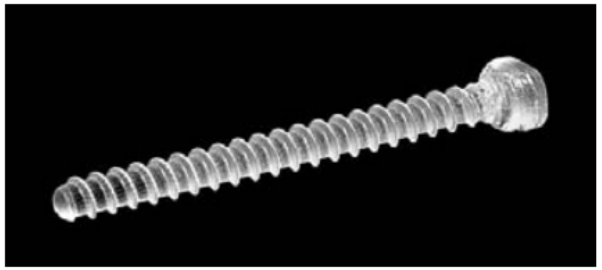
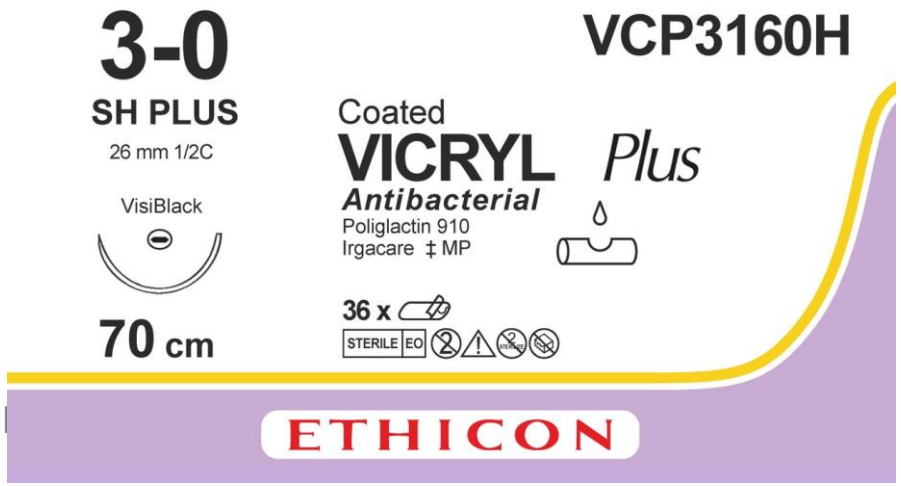


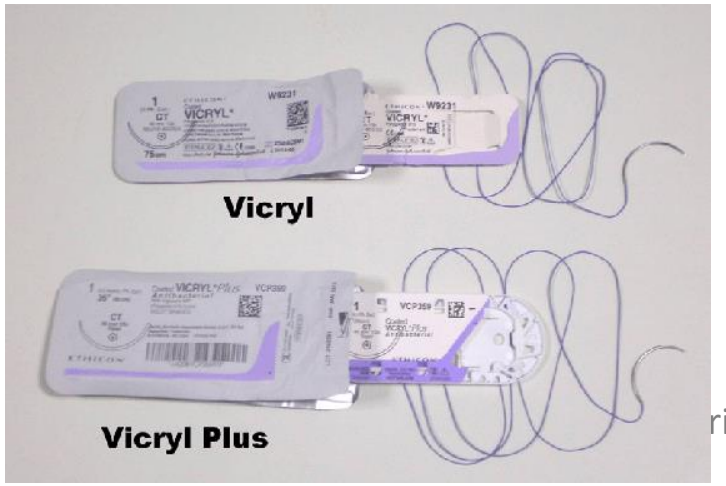
Figure 6.10

Example of a typical commercially available medical device made from polylactic acid and polyglycolic acid.

PLA/PGA Copolymers

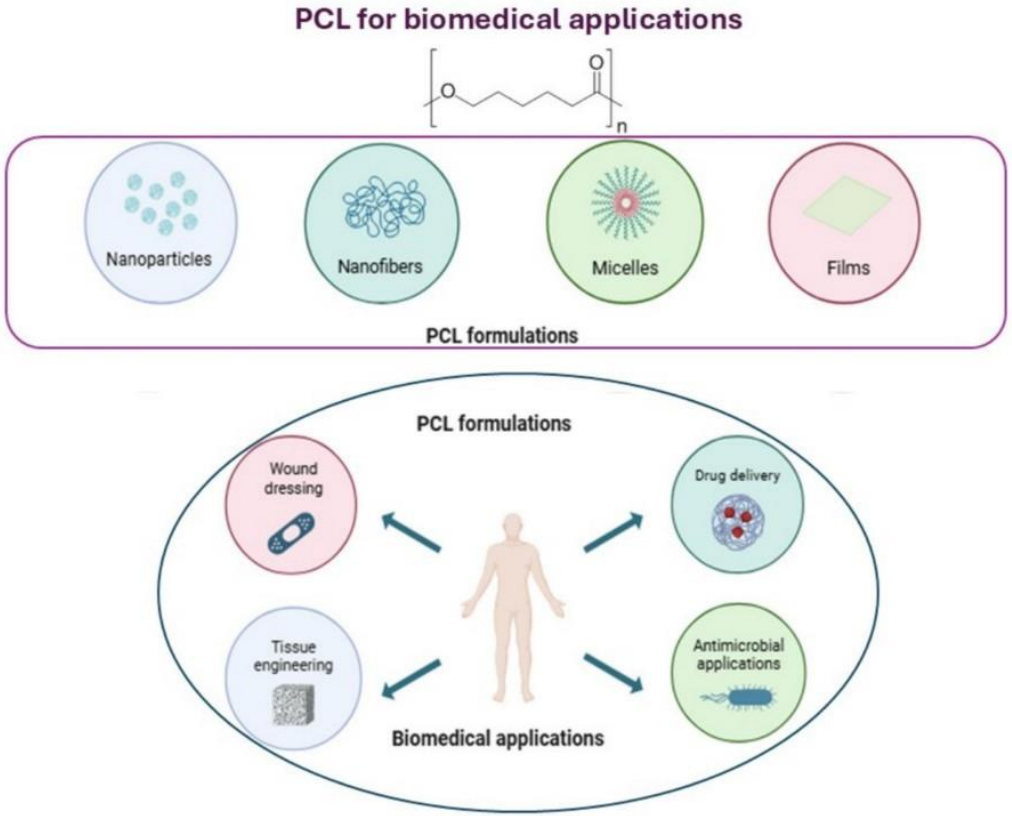
- PLA and PGA can be combined into copolymers with different ratios.
- Copolymer ratio affects degradation rate and mechanical behavior.
- **Vicryl** is **an absorbable suture** made from **PGA/PLA copolymer**.
- 50:50 PLA/PGA often degrades faster and is useful for scaffolds and drug deliver (PLGA)





4) Polycaprolactone (PCL)

- **PCL** is a **biodegradable polymer** used in **tissue engineering** and **drug delivery**.
- It has low melting temperature and very low glass transition temperature.
- It is more rubbery at room temperature compared with PLA.
- It degrades more slowly because it is hydrophobic and relatively crystalline.
- **PCL is useful for long-term drug delivery because it degrades slowly and can release drugs over extended periods.**

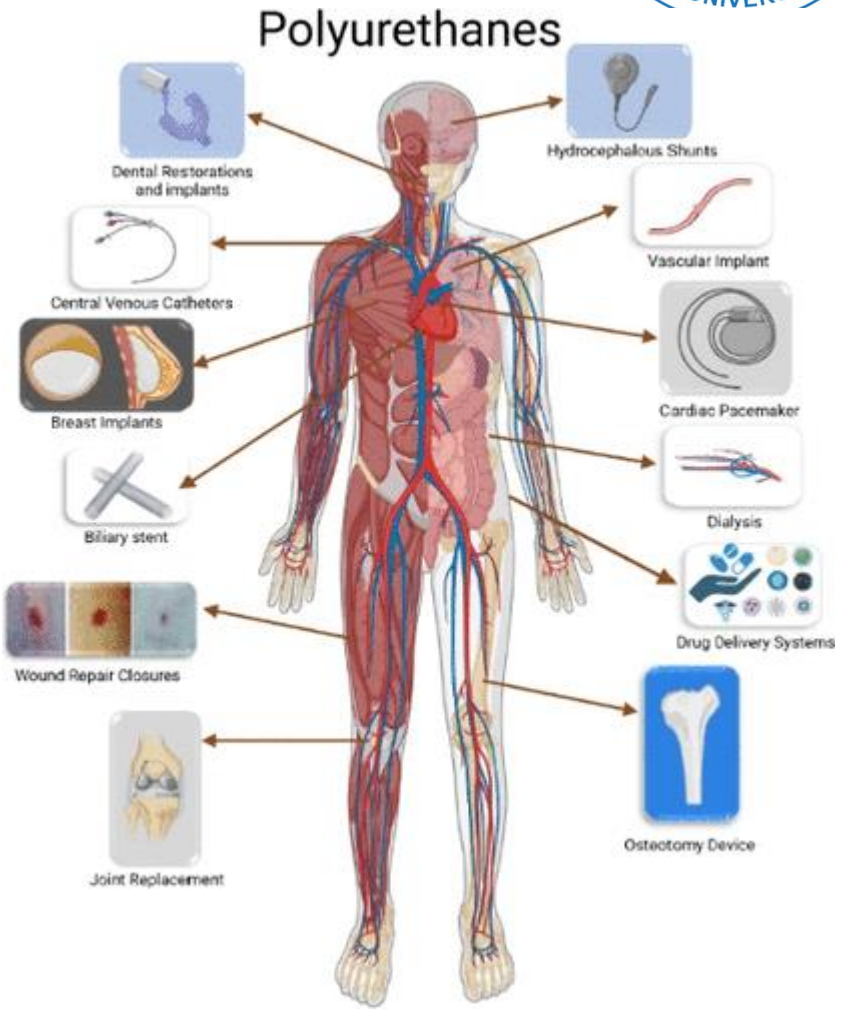


5) Other Biodegradable Polymers

- **Polyanhydrides** degrade mainly by surface erosion.
- They are useful for controlled drug delivery.
- They have been studied for drugs, proteins, growth factors, and chemotherapeutic agents.
- Other examples include **polyorthoesters**, **polycarbonates**, and **poly(p-dioxanone).**

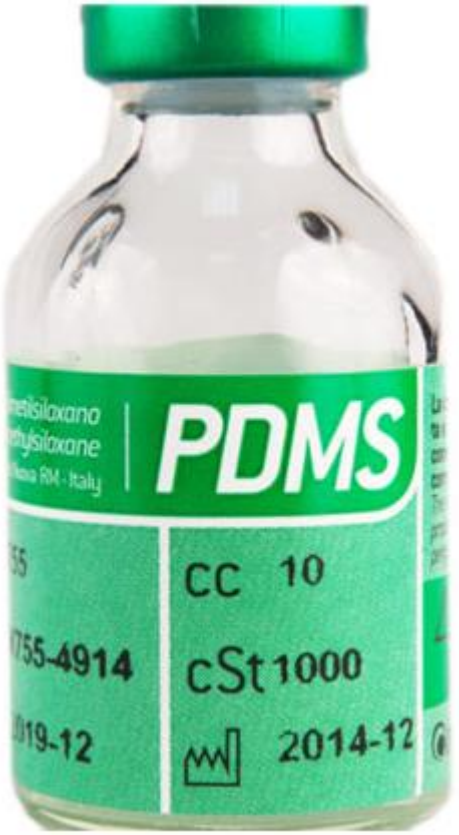
6) Polyurethanes

- Polyurethanes can contain both hard and soft segments.
- Their structure can be adjusted to produce soft or stiff materials.
- They have good mechanical properties and blood compatibility.
- Applications include **pacemaker leads**, **catheters, heart valves**, and **ligament reconstruction**.



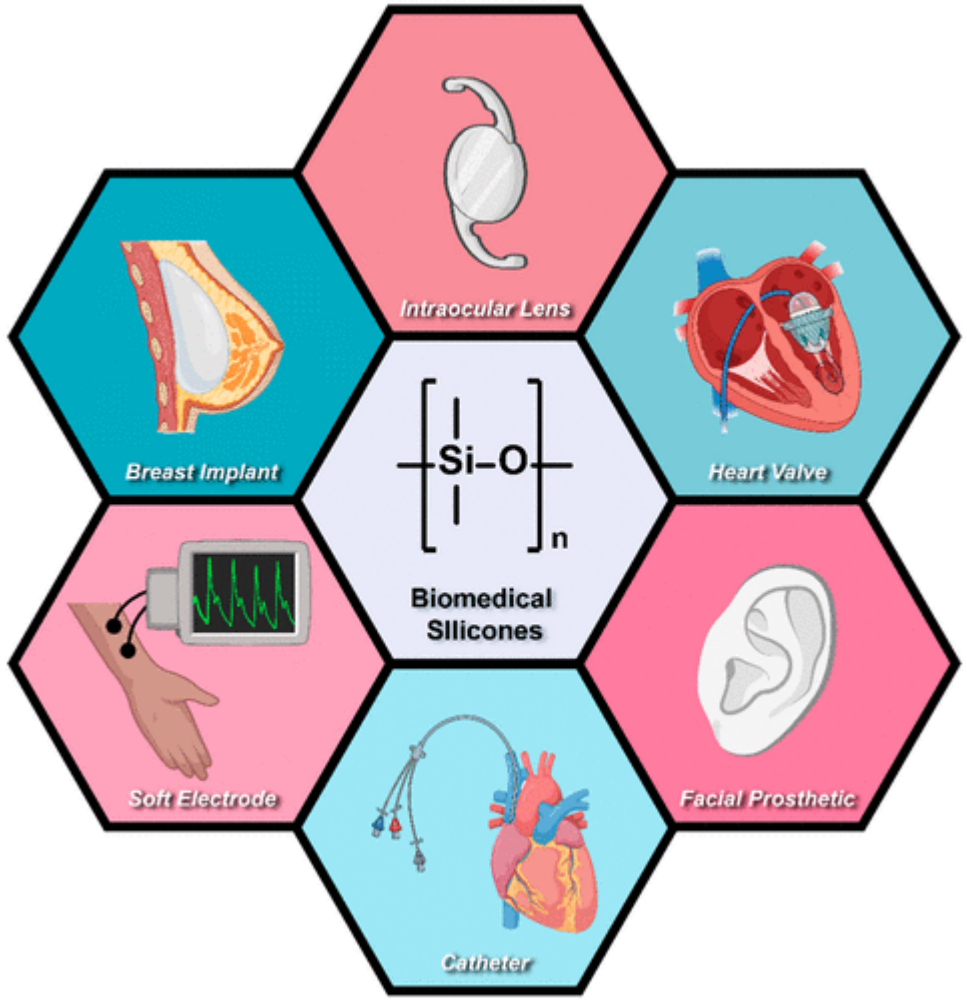
7) Silicones

- Silicones have alternating silicon and oxygen atoms in their backbone.
- The most common silicone is **PDMS**.
- They are flexible, hydrophobic, and gas permeable.
- Silicones can be crosslinked to form elastomeric networks.



Biomedical Applications of Silicones

- Used in catheters and cardiovascular devices.
- Used in finger joint, toe, and carpal bone prostheses.
- Used in breast reconstruction and augmentation implants.
- Also used in jaw, chin, and nasal support implants.



Khalid Abushahla, Doctoral Researcher in Biomedical Engineering

Summary of Common Polymeric Biomaterials

- **UHMWPE:** joint implant bearing surfaces, but wear debris is a concern.
- **PMMA:** bone cement, works mainly by mechanical interlocking.
- **PLA/PGA:** biodegradable implants, sutures, scaffolds, and drug delivery.
- **PCL:** slow-degrading polymer for tissue engineering and drug delivery.
- **Polyurethanes:** flexible medical devices and blood-contacting applications.
- **Silicones:** flexible elastomeric implants and catheters.

Hydrogels

Hydrogels as Biomaterials

- **Hydrogels** are **soft**, **jelly-like** polymeric materials.
- They contain hydrophilic polymer chains dispersed in water.
- They can **retain very high-water content**, sometimes **more than 99%.**
- Their **soft** and **hydrated** structure makes them useful in biomedical applications.





Classification of Hydrogels

- Hydrogels can be classified by source:
  - Natural hydrogels
  - Synthetic hydrogels
- They can also be classified by composition:
  - Homopolymer hydrogels
  - Copolymer hydrogels
  - Multipolymer hydrogels
  - Interpenetrating network hydrogels

Natural vs Synthetic Hydrogels

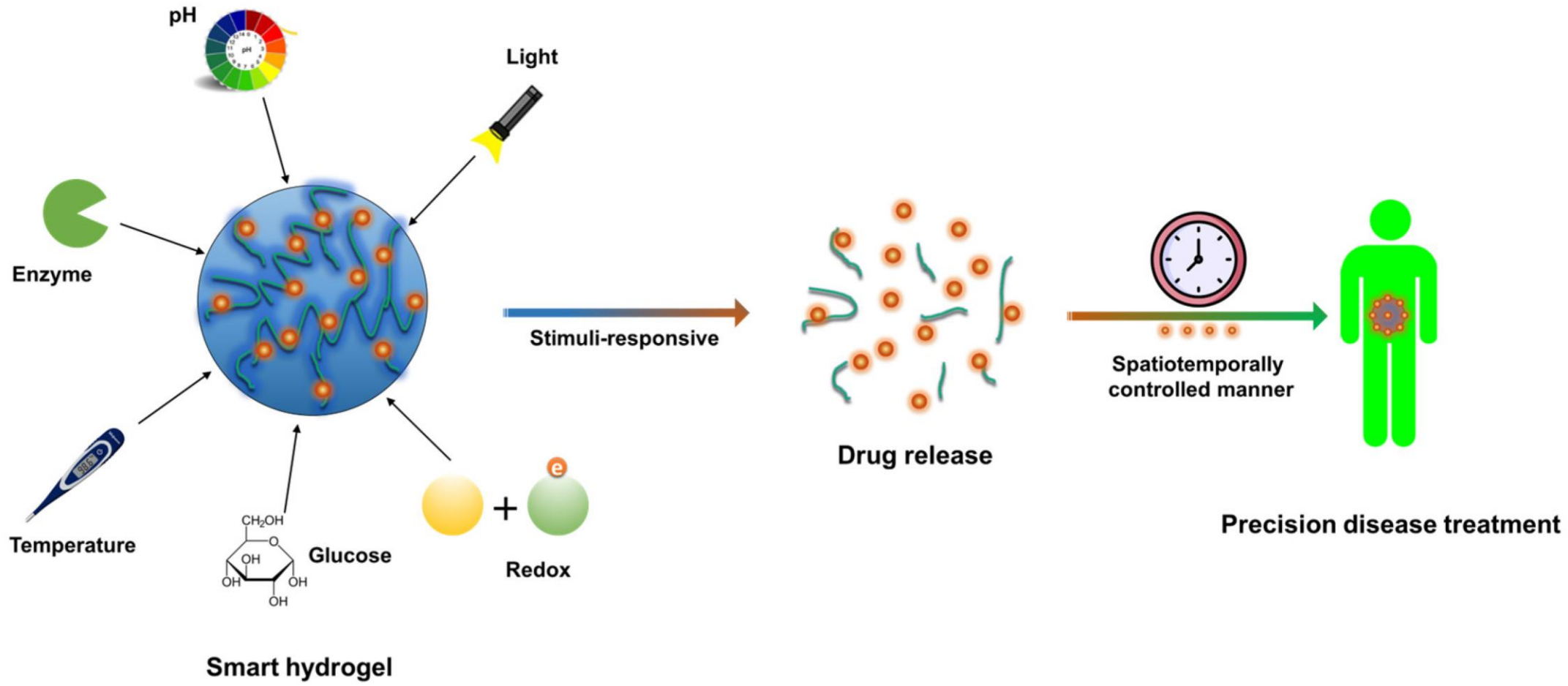
- **Natural hydrogels** include **chitosan**, **gelatin**, and **alginate**.
- Natural polymers may show variation in composition and properties.
- **Synthetic hydrogels** can be produced with more controlled structure.
- Synthetic hydrogels often have more consistent physicochemical properties.

Natural hydrogels	Synthetic hydrogels
Collagen	PolyHEMA
Gelatin	PEG / PEGDA
Alginate	PVA
Chitosan	PNIPAm
Hyaluronic acid	Polyacrylamide
Agarose	Polyacrylic acid
Fibrin	Polyurethane hydrogels
Cellulose-based hydrogels	Silicone hydrogels

Smart and Stimuli-Responsive Hydrogels

- **Smart hydrogels** respond to environmental changes.
- Common stimuli include:
  - Temperature
  - pH
  - Light
- They can change water affinity and switch between solution and gel forms.
- Useful for drug delivery and tissue engineering.

Smart and Stimuli-Responsive Hydrogels: Example (Drug release)



Smart Hydrogels: Temperature and pH Response

- **Smart hydrogels** respond to environmental changes such as **temperature**, **pH**, or **light**.
- **Temperature-responsive hydrogels:** swelling or gel structure changes when temperature changes.
  - Example: PNIPAm expands below its critical temperature and becomes more compact above it.
- **pH-responsive hydrogels:** contain acidic/basic groups that accept or release protons.
- These responses help control swelling, gel formation, and drug release.

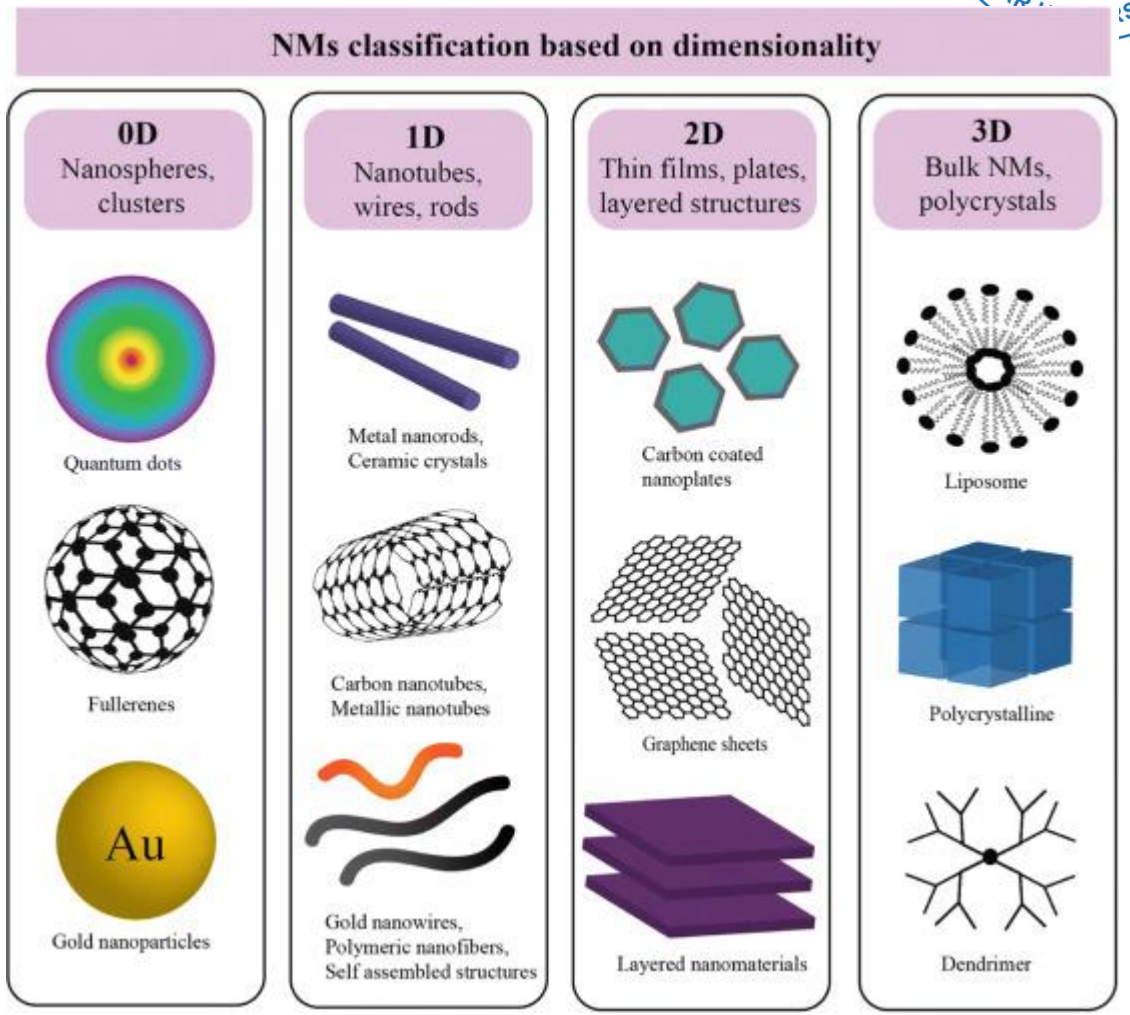
Biomedical Applications of Hydrogels

- **Contact lenses:** soft, comfortable, hydrated, and oxygen-permeable.
- **Drug delivery:** absorb water, swell, and release drugs gradually.
- **Tissue engineering:** act as soft scaffolds that can carry cells.
- **Wound healing:** keep wounds moist and support soft tissue repair.
- Their usefulness comes from **high water content**, **swelling ability**, **softness**, and **controllable release**.

Nanopolymers

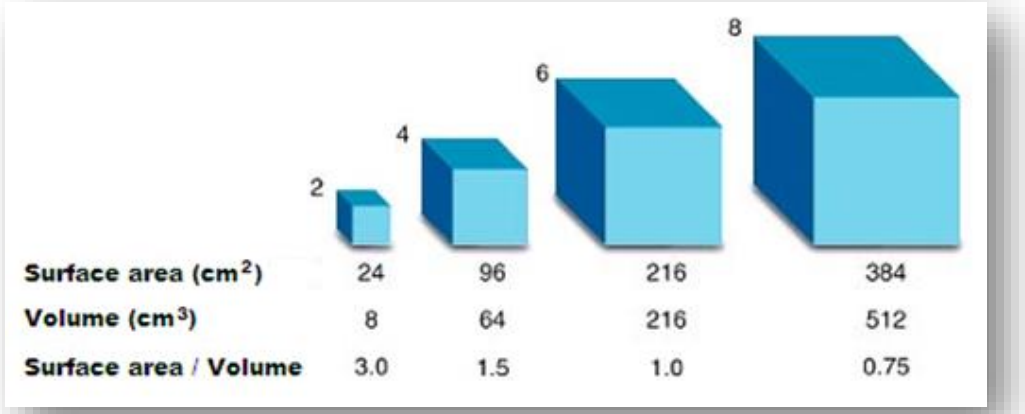
Nanopolymers

- Nanopolymers are polymer materials with at **least one dimension below 100 nm**.
- They can appear as **nanoparticles**, **nanofibers**, **nanotubes**, or **nanoscale coatings**.
- **Nanofibers** support **tissue scaffolds**, while **nanoparticles/nanotubes** can **carry and release drugs**.



Nanopolymers

- Their main advantage is very high surface area-to-volume ratio.
- This increases interaction with cells, tissues, drugs, and the surrounding environment.
- Their properties may differ from the same polymer in bulk form.



Surface area (cm^2)	24	96	216	384
Volume (cm^3)	8	64	216	512
Surface area / Volume	3.0	1.5	1.0	0.75

Summary of Polymeric Biomaterials

- Polymers are natural or synthetic long-chain materials used in many biomedical devices.
- Their properties depend on molecular structure, chain length, molecular weight, polymerization type, and physical state.
- Tg, Tm, crystallinity, crosslinking, and degradation behavior are key factors in polymer performance.
- Common biomedical polymers include UHMWPE, PMMA, PLA, PGA, PCL, polyurethanes, and silicones.
- Hydrogels are water-rich polymer networks used in contact lenses, drug delivery, tissue engineering, and wound healing.
- Nanopolymers provide high surface area and are useful for scaffolds, targeted drug delivery, and advanced therapies.

Think with me 🤔🤔🤔

- Explain the difference between monomer, oligomer, repeat unit, and polymer. Give one simple example.

- Compare addition polymerization and condensation polymerization. What is the main difference between them?

- Choose one biomedical polymer from the lecture, such as UHMWPE, PMMA, PLA/PGA, PCL, polyurethane, silicone, hydrogel, or nanopolymer, and explain its main biomedical use and why it is suitable for that use.



Thank you